

Abdul Rahim Memon

(902) 818-5652 | abdul.rhm@outlook.com | linkedin.com/in/abdul-rhm | github.com/abdrahim-123

OVERVIEW

Third-year Computer Science student with hands-on experience in data analysis, AI automation, and full-stack development. Seeking co-op opportunities to deliver impactful solutions at the intersection of data science and software engineering.

TECHNICAL SKILLS

Data & AI: Python (Pandas, NumPy, Scikit-learn), SQL, Llama 3 (LLM), R, Supabase (Vector DB)
Development: React.js, Node.js, C++, Java, HTML/CSS, Git, Docker, Linux/Bash
Tools & Systems: n8n (Workflow Automation), MongoDB, ESA SNAP, QGIS, REST APIs, TCP/IP
Soft Skills: Technical Communication, Strategic Planning, Adaptability, Problem Solving

PROFESSIONAL EXPERIENCE

Data Analysis Intern

June 2024 - July 2024

Unbox Robotics

Pune, India

- Optimized warehouse logistics models by implementing **Principal Component Analysis (PCA)**, reducing data redundancy and improving computational efficiency for predictive simulations.
- Developed a data-driven approach to parameter selection using **Hierarchical Clustering**, allowing the business to shift from manual estimations to statistical decision-making.
- Collaborated with engineering teams to streamline the data pre-processing pipeline, resulting in faster turnaround times for client simulations.

TECHNICAL PROJECTS & SYSTEMS

Intelligent AI Support Agent | *AI Automation & Workflow Optimization*

- Engineered an autonomous customer support system using **Llama 3** and **n8n**, capable of analyzing incoming ticket urgency and sentiment to prioritize critical business issues.
- Reduced response overhead by implementing **Retrieval-Augmented Generation (RAG)**, allowing the AI to access historical resolution data and draft accurate, context-aware responses automatically.
- Designed a dual-track architecture that separates routine inquiries from complex problems, optimizing resource allocation for support teams.

Shoreline Change Detection | *Environmental Analysis & Remote Sensing*

- Analyzed satellite imagery (RADARSAT) using **ESA SNAP** and **QGIS** to quantify coastal erosion trends in New Brunswick, providing actionable data for environmental monitoring.
- Executed a data processing pipeline including **Speckle Filtering** and **Radiometric Calibration** to transform raw sensor data into precise vector maps visualizing land-mass changes over a 4-year period (2019-2023).

Network Traffic & Security Pipeline | *Cybersecurity & Data Engineering*

- Developed a Python tool to capture and parse live network traffic, converting unstructured packet data into structured logs for security audits.
- Implemented real-time data capture using raw sockets, ensuring complete packet visibility for network diagnostics and anomaly detection.

Task Manager Pro | *Full-Stack Web Development*

- Developed Task Manager Pro, a **MERN Stack** application allowing users to manage tasks with real-time CRUD operations and persistent MongoDB storage.
- Implemented secure user authentication (Bcrypt/JWT) to ensure data privacy, allowing users to interact exclusively with their own records.

EDUCATION

Dalhousie University

Halifax, NS

Bachelor of Science in Computer Science – GPA: 4.18/4.30

Expected May 2027

- Relevant Coursework: Data Science (CSCI 3141), Intro to Software Projects (CSCI 2600), Intro to Database Systems (CSCI 2141)

Nirma University

Ahmedabad, India

Computer Science (First Two Years, Transfer) – CGPA: 9.32/10

2023 – 2025

- Awards: Academic Excellence and Scholar Awards
- Relevant Coursework: Probability and Statistics, Data Structures, Design and Analysis of Algorithms